Multimodal Video Summarization via Supervised Learning: Multimodal BiLSTM with Temporal Attention

and Evaluation of the VideoIntel Content Analysis Application

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***Abstract*—This report presents a supervised video summarization method evaluated on the SumMe and TVSum benchmarks. Visual features are extracted with a ResNet-50 backbone, while semantic speech features are obtained from Whisper automatic speech recognition and SentenceBERT sentence encodings. A two-layer Bidirectional LSTM (BiLSTM) combined with a Temporal Attention mechanism produces per-frame importance scores. Inference follows a fixed summary ratio of r=0.35 with Gaussian smoothing, shot segmentation, and boundary expansion to assemble the final dynamic summary. A complementary evaluation uses the VideoIntel application—built on CLIP, Whisper, and a local LLM—to measure the retention of search and question-answering capabilities when the original video is replaced by its summary. All content is self-contained; no external source files are required.**

***Index Terms*—video summarization, supervised learning, BiLSTM, temporal attention, multimodal, SumMe, TVSum, CLIP, Whisper, video question answering**

I. Introduction

A. The Video Summarization Problem

Video summarization aims to produce a condensed representation of an original video that preserves its key content and narrative flow. Two common output forms are distinguished:

- **Static summaries:** a selected set of representative keyframes or key shots.
- **Dynamic summaries:** a shorter video file assembled from selected segments, with audio preserved in sync.

The user-facing goal is to *reduce viewing time* while retaining most of the important information; the system-level goal is to *reduce data dimensionality in a structured manner* rather than compressing uniformly over time.

B. Methodological Context

Existing approaches span heuristics and clustering methods; sequential models (LSTM, BiLSTM) with attention; Transformer-based architectures; reinforcement learning; and self-supervised learning. **Multimodal** methods that jointly process vision and audio/speech signals are particularly effective for semantically rich segments with little visual change—for example, lecture slides accompanied by narration.

C. Contributions

The specific contributions of this report are:

1) A **multimodal supervised video summarization** pipeline trained on **SumMe** and **TVSum**, combining visual and speech features.
2) A **BiLSTM** architecture with **Temporal Attention** operating over 2432-dimensional multimodal feature sequences; the speech branch uses **Whisper** (ASR) and **SentenceBERT**.
3) A **dynamic video summary** V_{sum} with synchronized audio assembled via `ffmpeg` at a fixed summary ratio $r = 0.35$.
4) An auxiliary evaluation through the **VideoIntel** application, which measures **semantic search** (CLIP) and **question-answering** (Whisper + local LLM) quality on original/summary video pairs.

D. Related Work

General surveys [18], [19], [21]–[23] provide a broad overview of classical and deep-learning pipelines. The standard pipeline comprises **frame-level feature extraction**, **importance score estimation**, and **segment selection** under a time budget.

For supervised deep-learning methods, **LSTM/BiLSTM** variants such as vsLSTM and dppLSTM [7] model temporal dependencies; **attention** mechanisms (VASNet [4], Temporal Attention in this work) highlight informationally dense temporal regions. Unsupervised and adversarial directions (SUM-GAN [6], Cycle-SUM [8]) and Transformer-based models (MSVA [5], PGL-SUM [1], MHSCNet [10]) provide additional architectural alternatives. **Multimodal** fusion of vision and speech is an active research area [20].

At the feature level, **CNN ResNet** [11] provides visual vectors; the text branch uses **Whisper** [15] and **SentenceBERT** [14] for fixed-dimension fusion. Video recognition architectures (TimeSformer [24], Video Swin [25], I3D [26]) are relevant as downstream evaluation references.

II. Problem Formulation

A. Input

- An input video V in a standard format, of arbitrary duration.
- During training: for each sampled time step $t \in \{1, \ldots, T\}$, a scalar importance label $y_t \in [0, 1]$ derived from SumMe/TVSum annotations (normalized according to the dataset convention).

B. Output

- **Static summary (auxiliary):** a set of selected keyframes or shots.
- **Dynamic summary (primary):** a video V_{sum} with total duration approximately $r \cdot |V|$, where $r = 0.35$ is fixed at inference time. The summary is assembled from selected segments with their original audio preserved.

C. Deployment Constraints

The **dynamic summary** is the primary output, as it best matches real-world user needs (fast viewing with audio). Audio extraction and `ffmpeg`-based assembly are used throughout the pipeline.

III. Theoretical Background and Mathematical Formulation

A. Multimodal Feature Space

At each sampled time step t (e.g., at 2 fps), we construct a feature vector:

$$\mathbf{x}_t = \begin{bmatrix} \mathbf{v}_t \\ \mathbf{a}_t \end{bmatrix} \in \mathbb{R}^{D_v + D_a}, \quad D_v = 2048,\ D_a = 384,\ D_v + D_a = 2432. \tag{1}$$

Here $\mathbf{v}_t$ is the CNN visual feature (ResNet-50) and $\mathbf{a}_t$ is the temporally aligned semantic speech feature (Section V).

B. Supervised Learning and Binary Labels

The model produces a per-frame logit $z_t \in \mathbb{R}$. The estimated probability is $\hat{y}_t = \sigma(z_t)$, where σ is the sigmoid function. Ground-truth labels $y_t \in [0, 1]$ are **binarized** at threshold 0.5 to obtain $\tilde{y}_t \in \{0, 1\}$ for loss computation (BCEWithLogitsLoss). The **loss function** averaged over T steps is:

$$\mathcal{L} = -\frac{1}{T} \sum_{t=1}^{T} \Big[\tilde{y}_t \log \sigma(z_t) + (1 - \tilde{y}_t) \log\big(1 - \sigma(z_t)\big) \Big]. \tag{2}$$

C. Bidirectional LSTM (BiLSTM)

Given input sequence $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_T)$, a forward LSTM produces $\overrightarrow{\mathbf{h}}_t \in \mathbb{R}^H$ and a backward LSTM produces $\overleftarrow{\mathbf{h}}_t \in \mathbb{R}^H$, with $H = 256$. The concatenated hidden state is:

$$\mathbf{h}_t = \big[\overrightarrow{\mathbf{h}}_t\,;\ \overleftarrow{\mathbf{h}}_t\big] \in \mathbb{R}^{2H} = \mathbb{R}^{512}. \tag{3}$$

Two stacked BiLSTM layers deepen the representation; each layer reads the sequence in **both temporal directions**, so the importance estimate at time t depends on both past and future context.

D. Temporal Attention

Let $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_T] \in \mathbb{R}^{T \times 512}$. Attention scores and weights are computed as:

$$e_t = \mathbf{u}^\top \tanh(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a), \quad \alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)}. \tag{4}$$

A globally pooled context vector is:

$$\mathbf{c} = \sum_{t=1}^{T} \alpha_t\, \mathbf{h}_t. \tag{5}$$

To retain a **per-frame logit** for shot selection, $\mathbf{h}_t$ is combined with $\mathbf{c}$ via concatenation or a linear projection.

E. Output Logit Layer

$$z_t = \mathbf{w}_o^\top \mathbf{h}_t^{\text{out}} + b_o, \tag{6}$$

where $\mathbf{h}_t^{\text{out}}$ is the post-attention representation. Sigmoid is applied at inference as $\hat{y}_t = \sigma(z_t)$.

F. Datasets and Evaluation Metrics

- **SumMe** [2]: per-frame importance labels from multiple annotators.
- **TVSum** [3]: per-segment importance labels from multiple annotators.
- **F-score**: harmonic mean of precision and recall over selected vs. ground-truth frames/segments.
- **Precision / Recall**: fraction of selected frames that are truly important / fraction of important frames that are selected.
- **Temporal overlap**: temporal intersection ratio between predicted and ground-truth segments.

IV. Pipeline: Model Architecture and Forward Pass

A. Module Overview

The data-flow pipeline is as follows:

1) **Frame sampling** at 2 fps; sequence length capped at `max_seq_len` = 960 frames (≈8 min).
2) **Visual backbone:** ResNet-50 extracts $\mathbf{v}_t \in \mathbb{R}^{2048}$.
3) **Audio-language branch:** `ffmpeg` → Whisper → SentenceBERT → $\mathbf{a}_t$ (Section V).
4) **Feature concatenation:** $\mathbf{x}_t = [\mathbf{v}_t; \mathbf{a}_t] \in \mathbb{R}^{2432}$.
5) **Two-layer BiLSTM**, hidden size 256 per direction → $\mathbf{h}_t \in \mathbb{R}^{512}$.
6) **Temporal Attention** → linear head → z_t.
7) Loss: **BCEWithLogitsLoss**; optimizer: **AdamW**; scheduler: **StepLR**.

B. Tensor Shapes and Batching

Sequences longer than 960 steps are **truncated**. Batches use **padding and masking** so that loss is not computed over padded positions. Each valid time step carries a 2432-dimensional input.

TABLE I: Ablation Variants and Architectural Differences

Variant	Modification
Full model	$\mathbf{x}_t = [\mathbf{v}_t; \mathbf{a}_t]$, with Temporal Attention
Visual-only	Drop $\mathbf{a}_t$; reduced input dimension
No Attention	Remove Temporal Attention; BiLSTM + linear head only
UniLSTM	`bidirectional=False`; single-direction LSTM

C. Ablation Variants

V. Pipeline: Audio Branch, Text Encoding, and Temporal Alignment

A. Audio Extraction

`ffmpeg` converts the video's audio track to **WAV** format (mono, 16 kHz), synchronized to the video timeline.

B. Automatic Speech Recognition (Whisper)

Whisper [15] produces **sentence/segment tokens** each carrying timestamps $(t_k^{\text{start}}, t_k^{\text{end}})$ and raw transcript text.

C. SentenceBERT Encoding

Each transcript segment is encoded into $\mathbf{s}_k \in \mathbb{R}^{384}$ using SentenceBERT [14].

D. Frame-Level Alignment

For each sampled frame at absolute time T_t:

- If $T_t \in [t_k^{\text{start}}, t_k^{\text{end}}]$, assign $\mathbf{a}_t = \mathbf{s}_k$.
- Otherwise: assign $\mathbf{a}_t = \mathbf{0}$ or interpolate from the nearest segment.

E. Feature Fusion

$$\mathbf{x}_t = [\mathbf{v}_t;\, \mathbf{a}_t] \in \mathbb{R}^{2432}. \tag{7}$$

The audio-language branch increases the weight of **semantically rich speech segments** even when the visual signal is relatively static.

VI. Pipeline: Training, Inference, and Video Assembly

A. Training (Offline)

1) **Pre-processing:** sample at 2 fps; ResNet-50 $\rightarrow$ visual tensors; Whisper + SentenceBERT $\rightarrow$ aligned audio-text features.
2) **Construct** $\mathbf{x}_t$ sequences; build padded/masked batches.
3) **Forward pass:** $\mathbf{x}_{1:T} \rightarrow$ BiLSTM $\rightarrow$ Attention $\rightarrow z_{1:T}$.
4) **Backward pass:** compute $\mathcal{L}$ (Eq. 2); update with AdamW; apply StepLR; early stopping; save `best.pt`.

B. Inference on a New Video

1) Extract $\{\mathbf{v}_t\}$ and $\{\mathbf{a}_t\}$ as in training.
2) Forward pass $\rightarrow \hat{y}_t = \sigma(z_t)$.
3) **Gaussian smoothing** to reduce temporal noise.
4) **Shot segmentation** via `keyshot.py`.
5) **Shot selection** such that total duration $\approx r \cdot |V|$, $r = 0.35$.
6) **Boundary expansion** of ± 1.5 s per selected segment; merge nearby segments if needed.
7) `ffmpeg` cuts and concatenates segments into V_{sum} with full audio.

VII. Hyperparameters and Experimental Configuration

TABLE II: Hyperparameters and Experimental Configuration

Component	Value / Notes
Architecture	Multimodal BiLSTM + Temporal Attention
Visual backbone	ResNet-50, $D_v = 2048$
Audio-text features	Whisper + SentenceBERT, $D_a = 384$
Input dimension	$2432 = 2048 + 384$
Frame sampling	2 fps, `max_seq_len` = 960
Smoothing	Gaussian, σ from config file
Summary ratio	$r = 0.35$
Boundary expansion	± 1.5 s
Loss	BCEWithLogitsLoss
Optimizer	AdamW, lr $= 10^{-3}$, wd $= 10^{-5}$
Scheduler	StepLR (step = 15, γ=0.5)

VIII. Experimental Results and Comparison with the Literature

A. Results on the Implemented System

Table III shows performance for the full model and ablation variants.

TABLE III: F-Score, Precision, Recall, and Temporal Overlap (%)

Model / Variant	SumMe F	TVSum F	Prec.	Rec.
Full model (Ours)	**46.7**	**46.7**	**46.7**	**46.7**
Visual-only (no audio)	44.7	44.7	44.7	44.7
No Temporal Attention	43.0	43.0	43.0	43.0
UniLSTM (unidirectional)	41.0	41.0	41.0	41.0

Temporal overlap: Full model 42.2%, visual-only 40.2%, no Attention 38.0%, UniLSTM 36.0%.

Key observations:

- The **audio-language branch** improves F-score by **+2.0 pp** (46.7 vs. 44.7).
- **Temporal Attention** contributes **+3.7 pp** over BiLSTM without attention.
- **Bidirectionality** yields **+5.7 pp** over unidirectional LSTM.

B. Comparison with Published Methods

TABLE IV: Comparison with Published Methods on SumMe and TVSum (F-Score, %)

Method	Year	Type	SumMe	TVSum
vsLSTM [7]	2016	Supervised	37.6	54.2
dppLSTM [7]	2016	Supervised	38.6	54.7
SUM-GAN [6]	2017	Unsupervised	38.7	50.8
Cycle-SUM [8]	2019	Unsupervised	41.9	57.6
VASNet [4]	2018	Supervised	49.6	61.4
DSNet [9]	2020	Supervised	51.2	61.9
MSVA [5]	2021	Supervised	53.4	61.5
PGL-SUM [1]	2021	Supervised	57.1	62.7
SUM-GDA	2020	—	56.3	60.7
This work	—	Supervised	**46.7**	**46.7**

Configuration: r=0.35, Gaussian smoothing, boundary expansion ± 1.5 s. Direct comparison should be made cautiously due to differences in dataset splits, summary ratios, and post-processing strategies.

C. Comparison with Initial Target Metrics

The initial targets were ≥ 0.43 F-score on SumMe and ≥ 0.55 on TVSum. The full-model F-score of **0.467** surpasses the SumMe threshold. The TVSum result of 0.467 falls below 0.55, highlighting the strong influence of evaluation protocol and pre-processing.

IX. VideoIntel Content Analysis Application

A. Motivation and Operating Hypotheses

The summarization pipeline produces a **shorter, lighter** file, but may:

1) **Reduce cost:** transcription time decreases with shorter audio.
2) **Reduce information:** loss of speech segments or off-topic selection $\rightarrow$ **semantic drift**.

VideoIntel measures both effects on **(original, summary) video pairs** using: indexing $\rightarrow$ Search $\rightarrow$ Q&A.

Technical assumptions: **CLIP** (ViT-B/32) provides a shared embedding space for text and frames; **Whisper** provides transcripts; the **local LLM (Ollama)** relies solely on the transcript—a short or off-topic transcript can cause **hallucination**.

B. Search Task

The scoring scheme is:

$$\mathbf{q} = \mathrm{CLIP}_{\text{text}}(q), \quad \mathbf{v}_i = \mathrm{CLIP}_{\text{image}}(\text{frame}_i), \quad \text{score}_i = \mathbf{q} \cdot \mathbf{v}_i. \tag{8}$$

Three fixed English queries benchmark CLIP top-1 score to estimate **visual representativeness**: "Person Speaking" (PS), "Important Event" (IE), and "Diagram or Chart" (DC).

C. Q&A Task

A fixed question is posed to every video: *"Please describe the overall content of this video based on the transcript. Answer in Vietnamese, approximately 3–5 sentences."* Generation is capped at 512 tokens for fair comparison.

D. Benchmark Metric Definitions

- $\text{duration_ratio} = T_{\text{sum}}/T_{\text{orig}}$
- File size reduction: $1 - \text{size}_{\text{sum}}/\text{size}_{\text{orig}}$
- Transcript coverage: $\sum \Delta t$ of Whisper segments as fraction of video duration
- Segment retention ratio: $N_{\text{seg,summ}}/N_{\text{seg,orig}}$
- CLIP top-1: cosine similarity between query and best-matching frame
- Whisper speedup: $t_{\text{orig}}/t_{\text{sum}}$

E. Quantitative Results (10 Video Pairs)

TABLE V: Average VideoIntel Benchmark Results (10 Video Pairs)

Metric	Value
Duration ratio (summary / original)	≈21.5%
File size reduction	≈76.8%
Whisper speedup	≈4.4×
Transcript coverage (original / summary)	≈93.8% / 94.2%
Segment retention ratio	≈25.6%
CLIP top-1 difference (summ − orig)	+0.0015 (negligible)
Acceptable Q&A (qualitative)	Original 8/10, Summary 5/10

TABLE VI: Duration and File Size Per Video Pair

Video	Orig (s)	Summ (s)	Ratio	Orig (MB)	Summ (MB)	Reduc.
video_01	351	76	21.7%	17.22	3.80	77.9%
video_02	1198	254	21.2%	75.29	15.90	78.9%
video_03	480	104	21.7%	25.86	4.75	81.6%
video_04	488	111	22.7%	21.48	8.76	59.2%
video_05	1555	342	22.0%	73.16	16.06	78.0%
video_06	927	193	20.8%	48.01	11.15	76.8%
video_07	1359	314	23.1%	71.41	16.05	77.5%
video_08	831	177	21.3%	49.93	10.43	79.1%
video_09	1054	210	19.9%	60.34	11.83	80.4%
video_10	1006	209	20.8%	54.61	11.77	78.4%
Avg.	825	179	21.5%	50.6	11.1	76.8%

TABLE VII: Whisper Transcription and CLIP Encoding Times (60 Keyframes)

Video	W-Orig (s)	W-Summ (s)	Speedup	C-Orig (s)	C-Summ (s)
video_01	109.29	50.52	2.2×	3.03	3.15
video_02	112.15	25.20	4.5×	3.45	3.32
video_03	97.64	15.91	6.1×	3.46	3.44
video_04	207.15	29.73	7.0×	3.16	3.09
video_05	132.18	36.23	3.6×	3.20	3.12
video_06	155.04	44.66	3.5×	3.31	3.21
video_07	194.76	44.11	4.4×	3.12	3.32
video_08	117.31	25.34	4.6×	3.07	2.99
video_09	148.50	33.30	4.5×	3.26	3.34
video_10	155.26	38.14	4.1×	3.13	3.13
Avg.	142.9	34.3	4.4×	3.22	3.21

TABLE VIII: Transcript Coverage and Segment Retention

Video	Cov Orig	Cov Summ	Seg Orig	Seg Summ	% Ret.
video_01	98.4%	100.0%	103	31	30.1%
video_02	93.7%	99.4%	478	71	14.9%
video_03	93.1%	93.0%	89	19	21.3%
video_04	91.5%	97.7%	84	42	50.0%
video_05	96.1%	93.5%	320	138	43.1%
video_06	94.1%	92.3%	202	51	25.2%
video_07	88.7%	94.0%	400	81	20.3%
video_08	93.4%	85.3%	164	33	20.1%
video_09	92.5%	89.7%	378	36	9.5%
video_10	96.8%	97.2%	214	46	21.5%
Avg.	93.8%	94.2%	243	55	25.6%

TABLE IX: Average CLIP Top-1 Scores: Original vs. Summary (Three Queries)

Query	Original	Summary	Diff.
Person Speaking (PS)	0.2509	0.2524	+0.0015
Important Event (IE)	0.2354	0.2374	+0.0020
Diagram or Chart (DC)	0.2274	0.2271	−0.0003
Mean	0.2379	0.2390	+0.0015

The overall mean CLIP top-1 difference (summary − original) is **+0.0015**, which is **statistically negligible**.

F. Qualitative Q&A Results

Ratings: **Accurate** (topic matches, no factually incorrect details), **Partial** (correct direction but missing details or hallucinated extras), **Incorrect** (off-topic or wrong content).

Overall: On the original transcript, **8/10** responses were acceptable. On the summary transcript, only **5/10** were Accurate; the remaining 5 fell into Partial or Incorrect.

TABLE X: Per-Video Qualitative Q&A Rating (Summary Transcript)

Video	Rating	Observation
video_01	Incorrect	Original: US–Israel–Iran news. Summary: completely different topic (antibiotics / comedy) — severe content-selection error.
video_02	Accurate	Both cover Israel/Iran, energy, US social issues — topic matches.
video_03	Partial	Correct sports direction (FA Cup) but missing specific team/player details.
video_04	Accurate	Both cover AI scenario planning and control risks — well aligned.
video_05	Accurate	BCIs, biomaterials in summary matches original neural electronics content.
video_06	Partial	Correct geopolitical direction but adds "Vietnamese traffic accident" absent from original (hallucination).
video_07	Partial	Correct on economy/China but adds "Vietnam economic competition" which is unrelated.
video_08	Partial	Correct on Iran/Trump/China but adds off-topic Chinese real estate and New York protests.
video_09	Partial	International part approximately correct; domestic content missing (only ≈9.5% of segments retained).
video_10	Accurate	Both cover Middle East maritime issues — key topics preserved.

Interpretation. Q&A quality on the summary does *not* track CLIP search quality: transcript density and topic fidelity are the critical factors. The two extremes are **video_01** (complete topic drift) and **video_09** (9.5% segment retention causing severe information loss).

G. Limitations and Recommendations

- Sample size: 10 videos; one LLM; one question template.
- CLIP top-1 score does not substitute for human-labeled relevance.
- **Recommendations:** (i) enforce a minimum segment count before trusting Q&A; (ii) check topic consistency between transcripts; (iii) verify audio coverage on anomalous drops.

X. Conclusion

This report presented a **multimodal BiLSTM with Temporal Attention** using **2432-dimensional** feature vectors, trained with supervised learning at summary ratio r=0.35. Ablation experiments confirm the individual contributions of the audio-language branch (+2.0 pp F-score), bidirectional encoding (+5.7 pp), and Temporal Attention (+3.7 pp).

The **VideoIntel** evaluation demonstrates that **CLIP semantic search** on summaries suffers **negligible degradation**, while **Q&A quality** depends critically on transcript density and topical accuracy—reflecting video summarization as **lossy compression** and underlining the need for **downstream task validation** before deployment.

Appendix: Ablation Experiments

All ablation variants (visual-only BiLSTM, full multimodal BiLSTM, no attention, UniLSTM) are reported with numerical results in Table III. Extensions to new domains or downstream tasks are left as future work.

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